

Derivation of Short-Run Labor Supply Curve:

$$\text{Max } E \left[\sum_t \left(\frac{1}{1+p} \right)^t \{ (1-\phi) u(c_t) + \phi v(h_t) \} \right]$$

subject to,

$$\text{Budget constraint: } k_{t+1} = l_t w_t + (1+r_{t+1}) k_t - c_t$$

$$\text{Production function: } f(z_t, k_t, l_t) = z_t k_t^\alpha l_t^{1-\alpha}$$

$$\text{Labor endowment equation: } l_t + h_t = 1$$

$$\text{Stochastic shock process: } z_{t+1} = \phi z_t + \xi_{t+1}$$

~~$\therefore CL = E \left[\sum_t \left(\frac{1}{1+p} \right)^t \{ (1-\phi) u(c_t) + \phi v(h_t) \} \right]$~~

Assume log utility, then the objective function:

$$\text{Max } E \left[\sum_t \left(\frac{1}{1+p} \right)^t \{ (1-\phi) \log(c_t) + \phi \log(h_t) \} \right]$$

$$\therefore L = E \left[\sum_t \left(\frac{1}{1+p} \right)^t \{ (1-\phi) \log(c_t) + \phi \log(h_t) \} \right]$$

$$+ \lambda \left[l_t w_t + (1+r_{t+1}) k_t - c_t - k_{t+1} \right]$$

$$= E \left[\sum_t \left(\frac{1}{1+p} \right)^t \{ (1-\phi) \log(c_t) + \phi \log(h_t) \} \right]$$

$$+ \lambda \left[(1-h_t) w_t + (1+r_{t+1}) k_t - c_t - k_{t+1} \right]$$

$$\left[\because l_t + h_t = 1 \right]$$

FOC requires -

$$\frac{\partial L}{\partial c_t} = \left(\frac{1}{1+\rho} \right)^t \cdot \frac{(1-\theta)}{c_t} = \lambda_t \quad \dots (1)$$

$$\frac{\partial L}{\partial c_{t+1}} = E \left[\left(\frac{1}{1+\rho} \right)^{t+1} \cdot \frac{(1-\theta)}{c_{t+1}} \right] = \lambda_{t+1} \quad \dots (2)$$

$$\frac{\partial L}{\partial h_t} = \left(\frac{1}{1+\rho} \right)^t \cdot \frac{\theta}{h_t} = \lambda_t w_t \quad \dots (3)$$

$$\frac{\partial L}{\partial k_{t+1}} = \lambda_{t+1} (1+r_{t+1}) = \lambda_t \quad \dots (4)$$

Substituting (1) and (2) into (4)

$$E \left[\left(\frac{1}{1+\rho} \right)^{t+1} \cdot \frac{(1-\theta)}{c_{t+1}} \right] (1+r_{t+1}) = \left(\frac{1}{1+\rho} \right)^t \cdot \frac{(1-\theta)}{c_t}$$

$$\Rightarrow \left(\frac{1}{1+\rho} \right)^{t+1} \cdot (1-\theta) E \left[\frac{1}{c_{t+1}} \right] (1+r_{t+1}) = \left(\frac{1}{1+\rho} \right)^t \cdot \frac{(1-\theta)}{c_t}$$

$$\frac{\left(\frac{1}{1+\rho} \right)^{t+1}}{\left(\frac{1}{1+\rho} \right)^t} \cdot (1+r_{t+1}) = \frac{1}{c_t} E[c_{t+1}]$$

$$\Rightarrow \left(\frac{1}{1+\rho} \right)^{t+1-t} \cdot (1+r_{t+1}) = \frac{1}{c_t} \cdot E[c_{t+1}]$$

$$\Rightarrow \frac{\left(\frac{1}{1+\rho} \right)^{t+1}}{\left(\frac{1}{1+\rho} \right)^t} \cdot (1+r_{t+1}) \cdot E\left[\frac{1}{c_{t+1}} \right] = \frac{1}{c_t}$$

$$\Rightarrow \left(\frac{1}{1+\rho} \right)^{t+1-t} \cdot (1+r_{t+1}) \cdot E\left[\frac{1}{c_{t+1}} \right] = \frac{1}{c_t}$$

$$\Rightarrow \left(\frac{1}{1+\rho} \right) \cdot (1+r_{t+1}) \cdot E\left[\frac{1}{c_{t+1}} \right] = \frac{1}{c_t}$$

$$\Rightarrow \left(\frac{1+r_{t+1}}{1+\rho} \right) \cdot E\left[\frac{1}{c_{t+1}} \right] = \frac{1}{c_t}$$

$$\therefore \frac{1}{c_t} = \left(\frac{1+r_{t+1}}{1+\rho} \right) \cdot E\left[\frac{1}{c_{t+1}} \right]$$

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Here,

$$u(c_t) = \log(c_t)$$

and

$$u(c_{t+1}) = \log(c_{t+1})$$

$$\therefore u'(c_t) = \frac{1}{c_t}$$

(6)

$$\therefore u'(c_{t+1}) = \frac{1}{c_{t+1}}$$

~~and $v(h_t) = \log(h_t)$~~

~~$\therefore v'(h_t) = \frac{1}{h_t}$~~

(7)

Substituting (6) and (7) into (5)

$$u'(c_t) = \frac{(1+r_{t+1})}{(1+\rho)} E[u'(c_{t+1})]$$

(8)

(5) and (8) are the Euler equation.

General form

using log utility

Now, from (5),

Leaving aside uncertainty, we can write $E\left[\frac{1}{c_{t+1}}\right] = \frac{1}{c_{t+1}}$,

Then,

$$\frac{1}{c_t} = \left(\frac{1+r_{t+1}}{1+\rho} \right) \cdot \frac{1}{c_{t+1}}$$

$$\Rightarrow \frac{c_{t+1}}{c_t} = \frac{1+r_{t+1}}{1+\rho} \quad (9)$$

Putting (1) into (3),

$$\left(\frac{1}{1+\rho} \right)^t \cdot \frac{\phi}{h_t} = \left(\frac{1}{1+\rho} \right)^t \cdot \frac{(1-\phi)}{c_t} \cdot w_t$$

$$\Rightarrow \phi \cdot \frac{1}{h_t} = (1-\phi) \cdot \frac{1}{c_t} \cdot w_t \quad (10)$$

Here,

$$v(h_t) = \log(h_t)$$

$$\therefore v'(h_t) = \frac{1}{h_t} \quad (11)$$

Putting (6) and (11) into (10),

$$\phi \cdot v'(h_t) = (1-\phi) \cdot u'(c_t) \cdot w_t \quad (12)$$

(12) says that marginal cost of lost leisure = marginal benefit of additional consumption.

Now, from (10),

$$\phi \cdot c_t = \cancel{(1-\phi) \cdot h_t} w_t (1-\phi) \cdot h_t$$

$$\Rightarrow w_t (1-\phi) h_t = \phi c_t \quad \dots (13)$$

For period $t+1$, (13) becomes,

$$w_{t+1} (1-\phi) h_{t+1} = \phi c_{t+1} \quad \dots (14)$$

Dividing (14) by (13),

$$\frac{w_{t+1} (1-\phi) h_{t+1}}{w_t (1-\phi) h_t} = \frac{\phi c_{t+1}}{\phi c_t}$$

$$\Rightarrow \frac{w_{t+1}}{w_t} \cdot \frac{h_{t+1}}{h_t} = \frac{c_{t+1}}{c_t}$$

$$\Rightarrow \frac{w_{t+1}}{w_t}$$

$$\Rightarrow \frac{h_{t+1}}{h_t} = \frac{c_{t+1}}{c_t} \cdot \frac{w_t}{w_{t+1}}$$

$$\Rightarrow \frac{h_{t+1}}{h_t} = \left(\frac{1+r_{t+1}}{1+\rho} \right) \cdot \frac{w_t}{w_{t+1}}$$

Assume r is exogenous, i.e. $r_{t+1} = r$

so

$$\Rightarrow \frac{h_{t+1}}{h_t} = \left(\frac{1+r}{1+\rho} \right) \cdot \frac{w_t}{w_{t+1}}$$

$$\Rightarrow \frac{1-h_{t+1}}{1-h_t} = \frac{1+r}{1+\rho} \cdot \frac{w_t}{w_{t+1}} \quad (15)$$

(15) is the short-run labor supply curve.

Derivation of Long-Run Labor Supply:

Since consumption is constant at income level:

$$c_t = \cancel{w_t} h_t = \cancel{w_t} (1 - h_t) \quad \dots (16)$$

Putting (16) into (12),

$$\phi v'(h_t) = (1 - \phi) \cdot u'[w(1 - h_t)] \cdot w_t$$

$$\Rightarrow (1 - \phi) \cdot u'[w(1 - h_t)] = \frac{\phi v'(h_t)}{w_t} \quad \dots (17)$$

Putting (16) into (10),

$$\phi \cdot \frac{1}{h_t} = (1 - \phi) \cdot \frac{1}{w_t (1 - h_t)} \cdot w_t$$

$$\Rightarrow \phi \cdot \frac{1}{h_t} = (1 - \phi) \cdot \frac{1}{(1 - h_t)}$$

$$\Rightarrow \phi(1 - h_t) = (1 - \phi) h_t$$

$$\Rightarrow \phi - \phi h_t = h_t - \phi h_t$$

$$\therefore h_t = \phi \quad \dots (18)$$

~~Assume $\phi = \frac{2}{3}$, $h_t = \phi = \frac{2}{3}$~~

~~(18) says that leisure~~

$$\Rightarrow 1 - l_t = \phi$$

$$\Rightarrow l_t = 1 - \phi \quad \dots (19)$$

Assume $\phi = \frac{2}{3}$, $l_t = \phi = \frac{2}{3}$,

$$l_t = 1 - \phi = \frac{1}{3}$$

(18) and (19) says that labor and leisure is independent of wage level.

(19) is long-run labor supply curve.

The basic Mechanics/ Derivation of Long Run elasticity of Labor supply:

Assume $\frac{1+r}{1+\rho} = 1$, then from (9), $c_{t+1} = c_t$,
That is consumption is constant. In
both periods, the marginal utility of
consumption will also be constant, ~~and~~
i.e. $u'(c_t) = \lambda$ and $(1-\phi) \cdot \lambda = \lambda$ is assumed
[constant $\times$ constant = constant]

Then, From (12),

$$\phi \cdot v'(h_t) = \lambda w_t \dots (20)$$

assume general functional form of
 $v(h_t)$,

$$v(h_t) = \frac{\sigma}{1-\sigma} \cdot h^{\frac{\sigma-1}{\sigma}}$$

$$\begin{aligned} \therefore v'(h_t) &= \frac{\sigma}{1-\sigma} \cdot \frac{\sigma-1}{\sigma} \cdot h^{\frac{\sigma-1}{\sigma} - 1} \\ &= \frac{\sigma}{(1-\sigma)} \cdot \frac{-(1-\sigma)}{\sigma} \cdot h^{\frac{\sigma-1-\sigma}{\sigma}} \end{aligned}$$

$$\therefore v'(h_t) = -h^{-\frac{1}{\sigma}}$$

Since Marginal utility ~~cannot~~ cannot be negative, we ~~can~~ omit (-) sign.

$$\therefore v'(h_t) = \cancel{a} h^{-\frac{1}{\sigma}} \dots \textcircled{20} \textcircled{21}$$

Substituting $\textcircled{21}$ into $\textcircled{20}$,

$$\cancel{\frac{w}{\sigma}} \cdot h^{-\frac{1}{\sigma}} = \lambda w_t$$

$$\Rightarrow h^{-\frac{1}{\sigma}} = \frac{\lambda w_t}{\cancel{\sigma}}$$

$$\therefore h_t = \left(\frac{\lambda w_t}{\cancel{\sigma}} \right)^{-\sigma} \textcircled{22}$$

$$\Rightarrow 1 - l_t = \left(\frac{\lambda w_t}{\cancel{\sigma}} \right)^{-\sigma}$$

$$\Rightarrow \therefore l_t = 1 - \left(\frac{\lambda w_t}{\cancel{\sigma}} \right)^{-\sigma} \textcircled{23}$$

$\therefore$ Long-run elasticity of Labor supply,

$$\varepsilon_{l,w} = \frac{dl}{dw} \cdot \frac{w}{l}$$

$$= \frac{d}{dw} \left[1 - \left(\frac{\lambda w_t}{\sigma} \right)^{-\sigma} \right] \cdot \frac{w}{l}$$

$$= -(-\sigma) \cdot \left(\frac{\lambda w_t}{\sigma} \right)^{-\sigma-1} \cdot \frac{\lambda}{\sigma} \cdot \frac{w}{l}$$

$$= \sigma \cdot \left(\frac{\lambda w_t}{\sigma} \right)^{-\sigma-1} \cdot \left(\frac{\lambda w_t}{\sigma} \right) \cdot \frac{1}{l}$$

$$= \sigma \cdot \left(\frac{\lambda w_t}{\sigma} \right)^{-\sigma-1+1} \cdot \frac{1}{l_t}$$

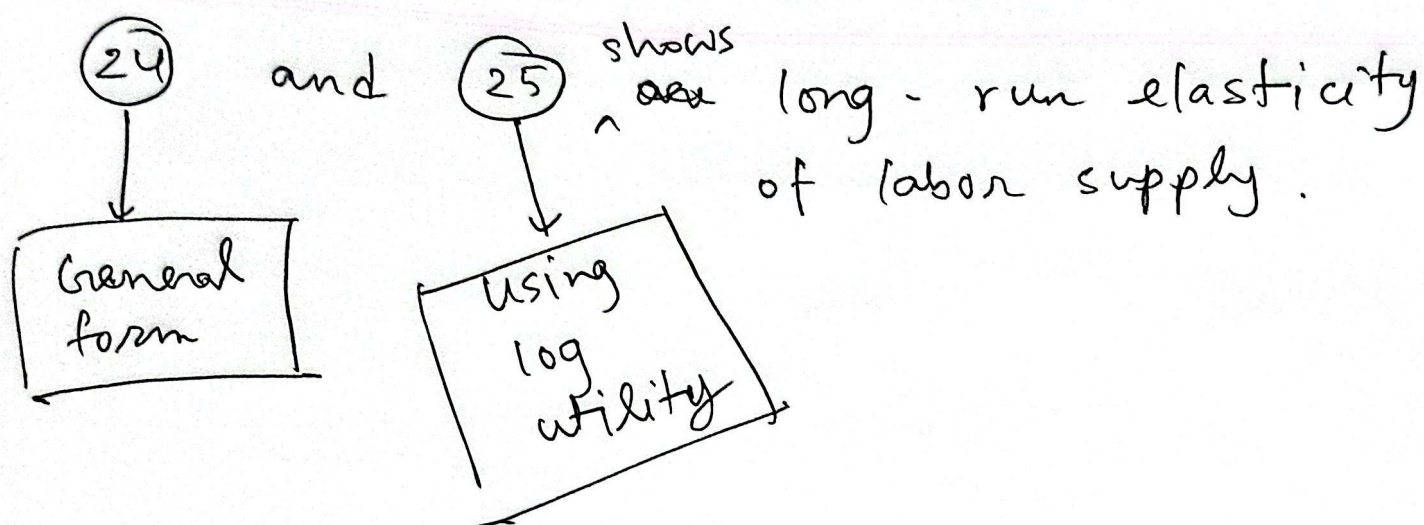
$$= \sigma \cdot \left(\frac{\lambda w_t}{\sigma} \right)^{-\sigma} \cdot \frac{1}{l_t}$$

$$= \sigma \cdot l_t \cdot \frac{1}{l_t} \quad \left[\text{From (22)} \right]$$

$$\therefore \varepsilon_{l,w} = \sigma \cdot \frac{l_t}{l_t} \quad \dots \quad (24)$$

If we assume log utility, $\sigma = 1$,

$$\text{Then } \varepsilon_{l,w} = \frac{l_t}{l_t} \quad \dots \quad (25)$$



Indivisible Labor Solution

$$= \frac{2\pi\omega}{\alpha}$$